

Week - 01

Class - 01

12.05.2025

Monday

Ordinary differential Equation

Math - 1213

$$y = e^{mx}$$

$$\frac{dy}{dx} = m e^{mx} = m y$$

Solve $\frac{dy}{dx} = m y$ O.D.E

$\left[\frac{d}{dx} \right] \rightarrow$ operation.

$$\frac{d^2}{dx^2} (\cos mx) = -m^2 \cos mx$$

$$\frac{d^2 y}{dx^2} = -m^2 y$$

$\hookrightarrow y = ?$

Answer: $y = \cos mx$

S.H.M

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \rightarrow \text{particular solution}$$

$$x = \cos \omega t$$

$$x = \sin \omega t$$

General, $x = \cos \omega t + \sin \omega t$

$$\frac{d^2 x}{dt^2} = -\omega^2 (\cos \omega t + \sin \omega t) = -\omega^2 x$$

Law of superposition

$$m^2 + 4 = 0 \quad \text{--- (1)}$$

$$\Rightarrow m^2 = -4$$

$$\Rightarrow m = \pm 2i$$

$$m = +2i, -2i$$

value

we are using superposition ans.

$$x = A \cos \omega t + B \sin \omega t.$$

$$\text{magnitude} = \sqrt{A^2 + B^2} \cos(\omega t + \phi)$$

$$\text{phase } \phi = \tan^{-1}\left(\frac{B}{A}\right).$$

$$\left(\frac{d}{dt} + i\omega\right) \left(\frac{d}{dt} - i\omega\right) x =$$

$$(D^2 + \omega^2)x = 0$$

$$\rightarrow D = \frac{d}{dt}$$

$$\frac{dN}{dt} = -kN$$

$$\rightarrow N = N e^{-kt}$$

$$\frac{dx}{dt} = i\omega x$$

$$x = A e^{i\omega t} + \bar{A} e^{-i\omega t}$$

conjugate

$$(D^2 + \omega^2)$$

$$x = A e^{i\omega t} + \bar{A} e^{-i\omega t}.$$

Week-02

class-01

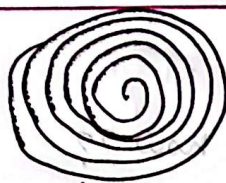
Math (NEW sine)??

19.5.25

Monday



Density



Density

Calculus:



differentiation



Integration

$$\frac{dN}{dt} = -iN$$

$$N = Ae^{-it}$$

$$e^{it} = \cos t + i \sin t$$

$$e^{-it} = \cos t - i \sin t$$

(+)

$$e^{it} + e^{-it} = 2 \cos t$$

$$\therefore \frac{e^{it} + e^{-it}}{2} = \cos t$$

$$\text{Similarly, } \frac{e^{it} - e^{-it}}{2} = \sin t$$

$$N = A \cos t + B \sin t$$

$$D = \frac{d}{dt}$$

$$\frac{74j}{5p} = \frac{74p}{5p}$$

$$0 = 11(j+1)$$

$$0 = 11(j-1)$$

$$0 = 11(j-1)(j+1)$$

$$0 = 11(1+1)$$

$$\frac{2j}{5} \leftarrow \frac{2p}{5}$$

$$(1+1) \frac{2p}{5}$$

$$1+1 \frac{2p}{5}$$

$$\frac{2j}{5} = (1+1) \frac{2p}{5}$$

same answer



22.2.01

problem

$$\frac{dN}{dt} = -iN \rightarrow \text{now??}$$

$$(D+i) N = 0$$

$$(D-i) N = 0$$

$$(D+i)(D-i) N = 0$$

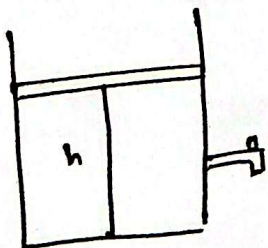
$$(D^2+1) N = 0$$

$$D^2 \rightarrow i^2 \rightarrow \text{amgvi}$$

$$\frac{d^2}{dt^2} (\cos t) = i^2 \cos t$$

$$\frac{d^2}{dt^2} (\cos t) = i^2 \Rightarrow \frac{e^{it}}{2} + (-1)^2 \frac{e^{-it}}{2}$$

$e \rightarrow$ natural base.



$h = h \cdot e^{-kt}$
 $\rightarrow$ amra top
 waktawa jita flow jani jara.
 jikata " " " "

$$\frac{dN}{dt} \propto N$$

$$m \frac{d^2 x}{dt^2} = -kx \quad ???$$

Week: 03

Class: 01

27.5.2025

Tuesday

$dx, dy \rightarrow$ differential.

$$m dx + N dy = 0.$$

$$dx + dy = 0.$$

Integrate $\rightarrow$
 $x + y = c$

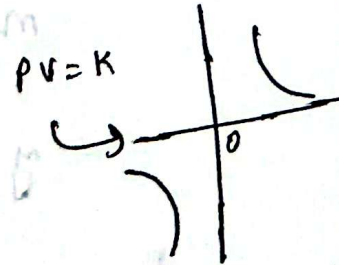
$$y dx + x dy = 0.$$

$$\frac{dx}{x} + \frac{dy}{y} = 0.$$

$$\text{Integrate} \rightarrow \ln x + \ln y = \ln c.$$

$$xy = c.$$

$$d(xy) = 0.$$



$$d\phi = M dx + N dy = 0$$

$$d\phi = \frac{\partial \phi}{\partial x} \cdot dx + \frac{\partial \phi}{\partial y} \cdot dy \rightarrow \text{partial differentiation}$$

$$\text{अतः, } M = \frac{\partial \phi}{\partial x}, N = \frac{\partial \phi}{\partial y} \rightarrow \text{प्रमाणित करें}$$

$$\frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x} \rightarrow \text{same but order}$$

$$\Rightarrow \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$$

$$\text{Ex. 1. } M = y$$

$$y dx + N dy = 0$$

अतः?

$$M = y, N = x$$

$$M = y, N = x$$

$$\therefore y dx + x dy = 0 \rightarrow \text{same}$$

$$\text{अतः, } y dx - x dy = 0 \rightarrow \text{अब अलग अलग करें}$$

$$\frac{y dx - x dy}{y^2} = 0 \quad \rightarrow \quad M = \frac{1}{y}, \quad N = -\frac{x}{y^2}$$

$$d\left(\frac{x}{y}\right) = 0 \quad \rightarrow \quad \frac{dN}{dx} = \frac{dM}{dy} \quad \text{exact?}$$

$$\frac{x}{y} = \left(\frac{1}{m}\right) \rightarrow \text{so } m = c \quad \text{exact}$$

$$x = cy$$

Integrating
function

$$\frac{dx}{x} - \frac{dy}{y} = 0 \quad \rightarrow \quad \ln x - \ln y = \ln m$$

$$\frac{x}{y} = \frac{1}{m}$$

so exact

$$(-x + y) dx + (x + y) dy = 0 \quad \rightarrow \quad \frac{dM}{dy} = 1, \quad \frac{dN}{dx} = 1$$

$$-x dx + y dx + x dy + y dy = c$$

$$d\left(-\frac{x^2}{2} + xy + \frac{y^2}{2}\right) = 0$$

$$\Rightarrow \boxed{-\frac{x^2}{2} + xy + \frac{y^2}{2} = c}$$

$c = 0$ pair of straight lines

pair of conics, etc.

4th week

class: 01

Energy form.

17.06.2025

$$\frac{1}{2} m v^2 + \frac{1}{2} k x^2 = E$$

$$\Rightarrow \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 + \frac{1}{2} k x^2 = E$$

differentiate.

$$\frac{1}{2} \cdot m \cdot 2 \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} + \frac{1}{2} \cdot 2 k x \cdot \frac{dx}{dt} = 0$$

$$\Rightarrow m \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} + k x \frac{dx}{dt} = 0$$

$$\Rightarrow m \frac{d^2x}{dt^2} + kx = 0 \quad \text{Differential form.}$$

Newton force
(force of inertia)

restoring force

2nd differential form (2nd Energy form)
 2nd (2nd) integration into 2nd
 function. or

2nd $\frac{dy}{dx} \rightarrow$ Integrating

2nd (2nd) Energy form $\rightarrow$ back into or.

$$m \frac{dn}{dt} \cdot \frac{dn^2}{dt^2} + k \cdot n \frac{dn}{dt} = 0$$

$$\Rightarrow m \int \frac{dn^2}{dt^2} \cdot \frac{dn}{dt} \cdot dt + k \int n \cdot \frac{dn}{dt} \cdot dt = 0$$

$\rightarrow$ Integrate into "2nd" or

$$\Rightarrow m \int \frac{dv}{dt} \cdot v \cdot dt + k \int n \cdot dn = 0$$

$$\Rightarrow m \int v \cdot dv + k \int n \cdot dn = 0$$

$$\Rightarrow m \cdot \frac{v^2}{2} + k \cdot \frac{n^2}{2} = E$$

$$\Rightarrow \frac{1}{2} m v^2 + \frac{1}{2} k n^2 = E$$

$$\textcircled{1} dQ = \boxed{dE} + \boxed{Pdv}$$

$$dQ = C_v dT + Pdv$$

$$\frac{dQ}{dT} = \frac{C_v dT}{T} + \frac{Pdv}{T}$$

book: The art of computer programming → Donald.

now, $PV = RT$.

$$T = \frac{PV}{R}$$

so,

$$ds = \frac{dQ}{T} = \frac{C_V dT}{T} + \frac{R dV}{V}$$

Integrable, (Integration now).

disordermen

set, $ds = 0$

$$ds = 0 \rightarrow \text{disordermen}$$

$$0 = \ln V + \ln T + \text{const}$$

$$0 = \ln V + \ln T + \text{const}$$

$$0 = \ln V + \ln T + \text{const}$$

$$Vbq + Tbq = 8k \quad (*)$$

$$Vbq + Tbq = 8k$$

$$\frac{Vbq}{T} + \frac{Tbq}{T} = \frac{8k}{T}$$

6th week:

class: 01

5/9/2022

Q Integrating factor

Q ① $\frac{dy}{dx} + p(x)y = q$ ①

$$\frac{dy}{y} + p dx = 0$$

$$\Rightarrow \ln y + \int p dx = C = \ln A$$

$$\Rightarrow \ln y + \ln e^{\int p dx} = \ln A$$
$$y e^{\int p dx} = A$$

Integrating factor.

① So, for solving $e^{\int p dx}$ multiply by

$$e^{\int p dx} \left(\frac{dy}{dx} + p(x)y \right) = q e^{\int p dx}$$

$$\frac{d}{dx} (y e^{\int p dx}) = q e^{\int p dx}$$

$$y e^{\int p dx} = \int q e^{\int p dx} dx + A$$

$$y = e^{-\int p dx} \int q e^{\int p dx} dx + A e^{-\int p dx}$$

$$\frac{dN}{dt} + kN = 0.$$

$$\hookrightarrow N = N_0 e^{-kt}$$

QWAT, $\frac{dN}{dt} + kN = 1$
 so, $e^{\int k dt} = e^{kt}$ (constant.)

$$N = e^{-kt} \int 1 \cdot e^{kt} dt + \boxed{A e^{-kt}}$$

Physics. (यहाँ करें.)

$$= \boxed{\frac{1}{k}} + A e^{-kt}.$$

Inductance voltage.

$$L \frac{di}{dt} + Ri = E$$

Voltage drop.

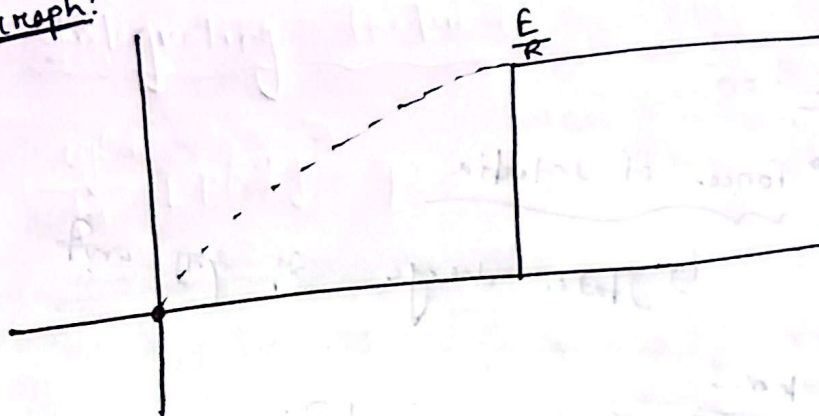
Voltage generator

$$\boxed{i = \frac{E}{R} + A e^{-\frac{R}{L}t}}$$

$\underline{i=0}$

$$0 = \frac{E}{R} + A e^{-\frac{R}{L}t} \left(1 - e^{-\frac{R}{L}t}\right)$$

Graph:



circuit ~~is~~ ^{is} not on the way ^{to} ^{the} ^{right}

or $\frac{E}{R} \sin \theta$

પણ, $t = 0$

$$N = N_0$$

$$N = ?$$

$N = ?$

यदि $u = v$ हो, right side के '1' को एक बार अवकलित करके

integrating factor निकालेंगे।

यदि $u \neq v$ हो, तो

$$\frac{dy}{dx} + 3x^2y = x^2$$

$$e^{\int 3x^2 dx} = e^{x^3} \quad (I, f)$$

2
विधि calculation का (नए) शर्त

C.T. ⑥ ઝેર વિશિષ્ટ
પ્રશ્ન.

$$m \frac{d^2 x}{dt^2} + kx = 0.$$

force of inertia

state change & 2π unit.

$$m \frac{d^2 x}{dt^2} = -kx \rightarrow \text{from hook's law.}$$

or, $x = \cos \omega t / \sin \omega t.$

$$\rightarrow (m\omega^2 - k) \cos \omega t + k \cos \omega t = 0$$

$$\Rightarrow (m\omega^2 - k) + k = 0$$

$$\Rightarrow (m\omega^2 - k) \sin \omega t + k \sin \omega t = 0$$

$$\omega = \sqrt{\frac{k}{m}}$$

angular frequency

$$\text{or, } x = \cos \sqrt{\frac{k}{m}} t, \sin \sqrt{\frac{k}{m}} t.$$

Law of superposition:

$$x = A \cos \sqrt{\frac{k}{m}} t + B \sin \sqrt{\frac{k}{m}} t$$

$$= a \cos \left(\sqrt{\frac{k}{m}} t - \phi \right)$$

amplitude

phase

$$\left| \begin{array}{l} A = a \cos \phi \\ B = a \sin \phi \end{array} \right|$$

$$x = a, \quad t = 0, \quad \frac{dx}{dt} = 0$$

$$\phi = \tan^{-1}\left(\frac{B}{A}\right)$$

~~$$x = 0, \quad t = 0, \quad \frac{dx}{dt} = 0$$~~

$$\text{Now, } x = a \cos\left(\sqrt{\frac{k}{m}} t - \phi\right)$$

$$\text{differentiate } \cos(\omega t), \sin(-\phi) = 0$$

$$\phi = 0$$

$$\text{Now, } x = a \cos\sqrt{\frac{k}{m}} t$$

$$\text{Now, } x = 0, \quad t = 0, \quad \frac{dx}{dt} = 0$$

$$x = a \cos\left(\sqrt{\frac{k}{m}} t - \phi\right)$$

$$0 = a \cos(-\phi)$$

$$\text{Now, } \phi = \frac{\pi}{2}$$

$$\therefore x = a \sin\sqrt{\frac{k}{m}} t$$

Week: 08

Class: 01

26/09/2020

RLC circuit:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E$$

Interference
force

Capacitance
force.

$$m \frac{d^2 x}{dt^2} + L \frac{dx}{dt} + Kx = 0$$

shm damping force potential f

$$\rightarrow L \frac{d^2 q}{dt^2} + 0 + \frac{1}{LC} q = 0 \quad \text{--- (1)}$$

$$q = A \cos \sqrt{\frac{1}{LC}} t + B \sin \sqrt{\frac{1}{LC}} t$$

① we know $q \frac{dq}{dt}$ is the work done

$$\frac{1}{2} L \left(\frac{dq}{dt} \right)^2 + \frac{1}{2C} q^2 = E$$

$$q = A \cos \sqrt{\frac{1}{LC}} t$$

when $t = 0$

$q = A \rightarrow \text{max (Capacitance \& energy only)}$

when $\frac{1}{\sqrt{LC}} t = \frac{\pi}{2}$

$q = 0 \rightarrow \text{min (Capacitance Inductance \& energy only)}$

auxiliary equation:

$$\frac{d^2 x}{dt^2} + x = 0$$

$$x = e^{\lambda t}$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

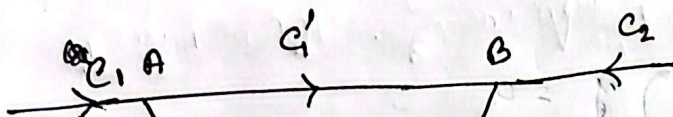
$$e^{it} = \cos t + i \sin t$$

$$x = A e^{it} + A e^{-it}$$

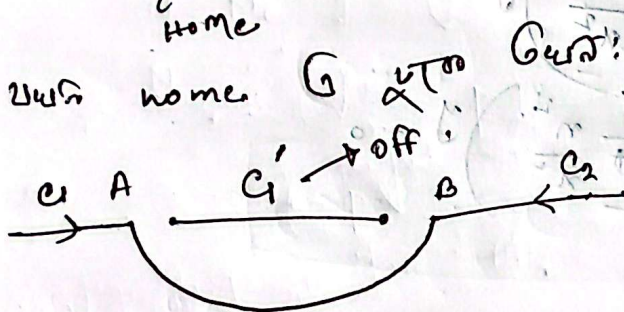
$$= \left(\frac{a - ib}{2} \right) (\cos t + i \sin t)$$

$$+ \left(\frac{a + ib}{2} \right) (\cos t - i \sin t)$$

$$x = a \cos t + b \sin t$$



LIFO



$$dx = x_2 - x_1$$

LIFO

Question

Legendre equation: differential equation:

$$\rightarrow (1-x^2)y'' - 2xy' + \lambda y = 0.$$

$$\text{Given } (1-x^2)y'' - 2xy' + \lambda y = 0$$

SHM

$\Rightarrow$ fine & cos curve.

Bessel's equation:

$$x^2 y'' + xy' + (x^2 - n^2)y = 0$$

$$\Rightarrow y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0.$$

$$\text{Given, } y'' + \frac{1}{x}y' + \left(1 - \frac{n^2}{x^2}\right)y = 0$$

Legendre's equation:

$$(1-x^2)y'' - 2xy' + \lambda y = 0$$

$$y = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$$

$$y' = a_1 + \dots + (n-1)a_{n-1}x^{n-2} + na_nx^{n-1}$$

$$y'' = \dots + (n-1)(n-2)a_{n-1}x^{n-3} + n(n-1)a_nx^{n-2}$$

$$(1-x^2) \left(\dots + (n-1)(n-2)a_{n-1}x^{n-3} + n(n-1)a_nx^{n-2} - 2x(a_1 + \dots + (n-1)a_{n-1}x^{n-2} + na_nx^{n-1}) + \lambda(a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n) \right) = 0$$

x^n coefficient

$$x^n \Rightarrow n(n-1)a_n - 2na_n + \lambda na_n = 0.$$

$$\Rightarrow n(n-1) - 2n + \lambda n = 0.$$

$$\therefore \lambda = n(n-1) + 2n = n(n+1)$$

$$x^{n-1} \rightarrow - (n-1)(n-2)a_{n-1} - 2(n-1)a_n + n(n+1)a_{n-1} = 0.$$

$$a_{n-1} = 0.$$

$$n=2$$

$$(1-x^2)y'' + 2xy' + 6y = 0$$

$$(1-x^2)x'(2a_2) - x \times 2a_1x + 6(a_0 + a_2x^2) = 0$$

$$\frac{x^2}{x^2} \rightarrow -2a_2 - 4a_2 + 6a_2 = 0$$

Let $n=2$.

Let $y = a_0 + a_2x^2$.

$$y = a_0 + a_2x^2.$$

x^0

$$2a^2 + 6a_0 = 0$$

$$\therefore a_2 = -3a_0$$

$$y = a_0 - 3a_0x^2$$

$$= a_0(1 - 3x^2)$$

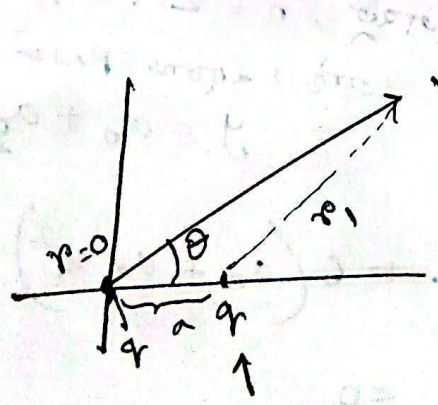
$$y(1) = 1 = a_0(-2)$$

$$a_0 = -1/2$$

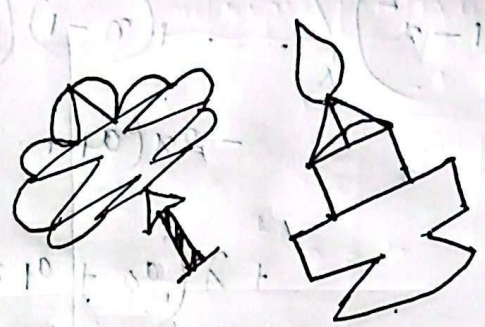
$$\therefore y = -1/2 (1 - 3x^2)$$

$$y = \frac{1}{2}(3x^2 - 1) = P_2(x)$$

ans: (v) $P_3(x)$ to ω_0 $\rightarrow$ Assignment.



$$\begin{aligned} r_1^2 &= r^2 + a^2 - 2ra \cos \theta \\ &= r^2 \left(1 - 2 \frac{a}{r} \cos \theta + \left(\frac{a}{r} \right)^2 \right) \\ &= r^2 (1 - 2xz + z^2) \end{aligned}$$



$$\sigma_2(n), v = \frac{dr}{r_1} = \frac{dr}{r\sqrt{1-2xz+z^2}}$$

$$\left(\frac{1-2xz+z^2}{x} \right)^{-1/2} = 1 + \frac{1}{2} (2xz - z^2) + \frac{3}{8} (2xz - z^2)^2$$

$$z^L = \frac{3}{2} (3x^2 - 1)$$

$$\begin{aligned} &= \frac{1}{\sqrt{1-x^2}}^{-1/2} \\ &= 1 + \frac{1}{2}x + 3 \end{aligned}$$

$$\left(\frac{1-2xz+z^L}{x} \right)^{-1/2}$$

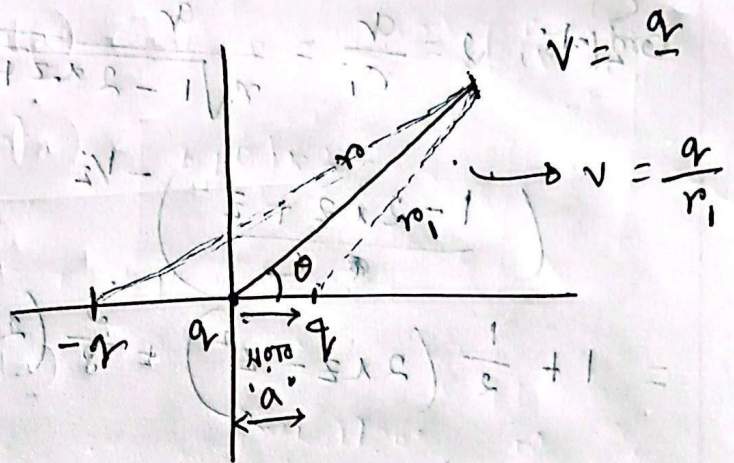
$$\text{generating function} = \sum p_n(x) z^n$$

$$x = 3 \times 4$$

$$y = 12$$

$$n = 3 \cdot 2(n)$$

C. T. $\rightarrow$ Legendary.



$$r_1^2 = r^2 + a^2 - 2ra \cos \theta$$

$$= r^2 \left(1 - 2 \cos \theta \left(\frac{a}{r} \right) + \left(\frac{a}{r} \right)^2 \right)$$

$$= r^2 \left(1 - 2xz + z^2 \right)$$

$$V = \frac{q}{r_1} = \frac{q}{r \sqrt{1 - 2xz + z^2}}$$

$$= \frac{q}{r} \left(1 - (2xz - z^2) \right)^{-1/2}$$

$$= \frac{q}{r} \left(1 + \frac{1}{2} (2xz - z^2) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!} (2xz - z^2)^2 + \dots \right)$$

$$= \frac{q}{r} \left(1 + xz + \left(-\frac{1}{2} + \frac{3}{8} 4x^2 \right) z^2 + \dots \right)$$

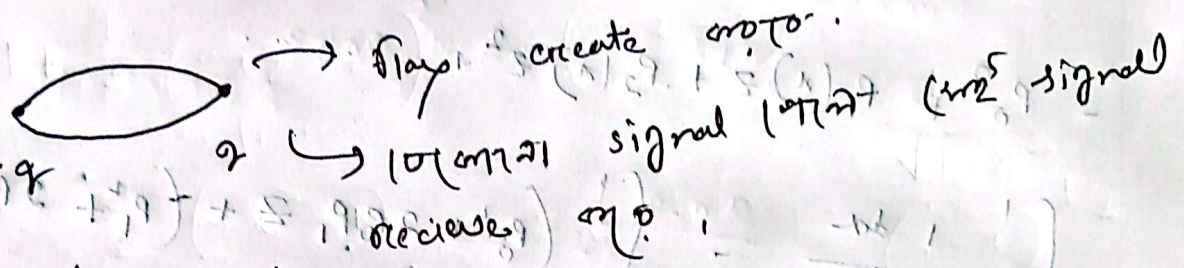
$$= \frac{q}{r} \left(\frac{p_0}{1} + \frac{p_1}{1} xz + \frac{1}{2} (3x^2 - 1) z^2 + \dots \right)$$

$$= \frac{q}{r} \sum p_n(x) z^n$$

$$= \frac{a}{n} \sum p_n (\cos \theta) \left(\frac{a}{n}\right)^n$$

→ -q point

$$- \frac{a}{n} \sum p_n \cos(\pi - \theta) \left(\frac{a}{n}\right)^n$$



Generating function:

$$\sqrt{1 - 2xz + z^2}$$

$$\frac{1 - 2xz + z^2}{1 - 2xz + z^2}$$

$$S = \frac{D}{\lambda c} \rightarrow \text{in-dec}$$

$$S = \frac{100}{\lambda c} \rightarrow \text{in-cm}$$

Laurent's half shade ~~to~~ polarimeter.

dextrorotatory $\rightarrow$ specific rotation of a compound is positive. The substances that rotate

levorotatory

~~transmission~~
frequency

$$c = \frac{m}{V}$$

Orthogonal Condition

Orthogonal function
Bessel's equation
Legendre's equation

with 0(n)?

exam 2

$\theta < 4^\circ \rightarrow$ SHM.
maximum kinetic energy $\rightarrow$ frequency (on wavelength) of the incident light & the work function of the metal.

$$\frac{d^2 x}{dt^2} + 2c \frac{dx}{dt} + \omega^2 x = \sin t$$

$$= e^{-t} \sin t$$

Bessel's equation:

$$y = (a_0 + a_1 x + \dots) x^s$$

Probation method

$$= \sum a_n x^{n+s}$$

→ x^s extra

$$a_{n+1} = -\frac{a_{n-2}}{(n+s)^2}$$

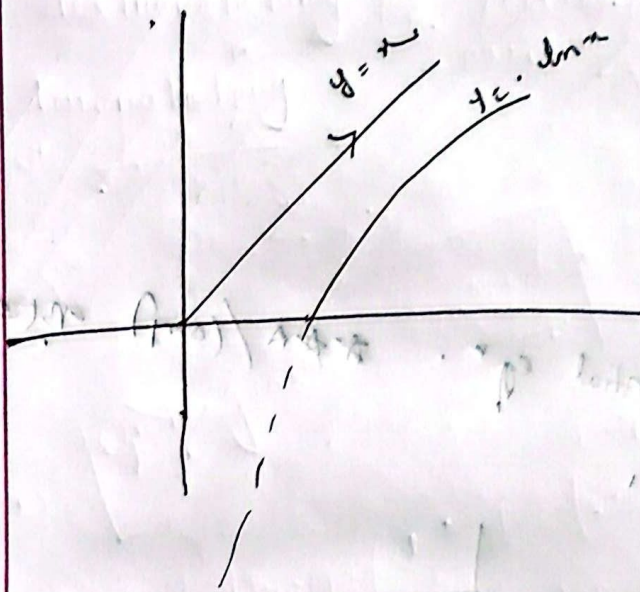
$$y = a_0 \left(1 - \frac{x^2}{(2+s)^2} + \frac{x^4}{(2+s)^2(4+s)^2} - \dots \right) x^s$$

partial differentiation

$$\left. \frac{\partial y}{\partial s} \right|_{s=0} =$$

$$\frac{d}{dx} (a^x) = a^x \ln a$$

$$\frac{d}{dx} (x^s) = x^s \ln x + \frac{x^s}{x}$$



$$\frac{\ln x}{x=0} \rightarrow \infty$$

exam.

Q1

$$\frac{dy}{dx} + y = y^3 \rightarrow \text{Bernoulli eqn.}$$

$$\frac{dy}{dx} + y = y^n \rightarrow n = 1, 2, 3, \dots$$

$$y^3 \frac{dy}{dx} + y^{-2} = 1$$

$$\text{let, } y^{-2} = v$$

$$\Rightarrow -2 y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\rightarrow -\frac{1}{2} \frac{dv}{dx} + v = 1$$

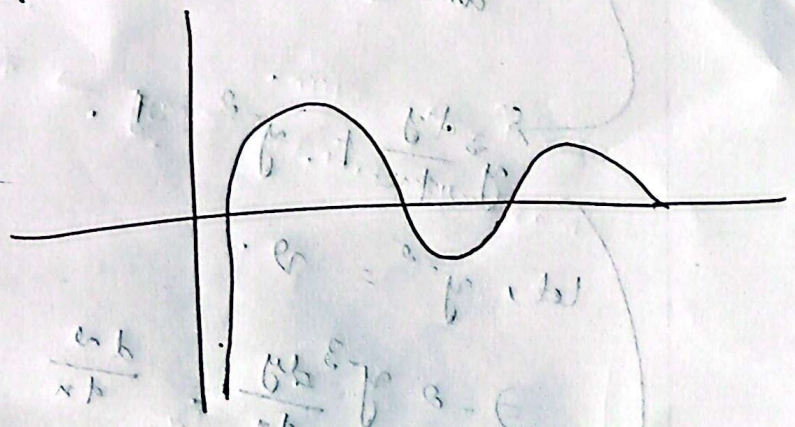
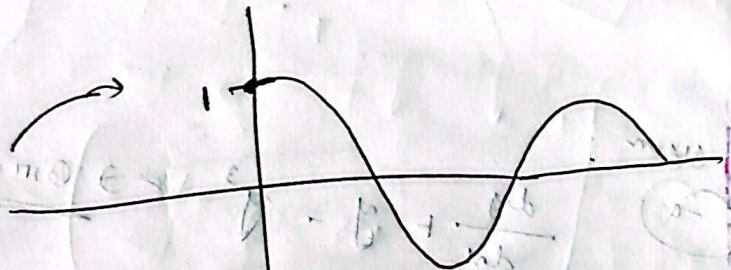
$$\frac{dy}{dx} + py = Q$$

$$Mdx + Ndy = 0$$

- y Second order Differential eqⁿ. ***A (comp nsto 2to)
- y Legendre's equation
- y Bessel's equation

$$1 - \frac{x^2}{2^2} + \dots$$

cos 2x is mth order
sin 2x is mth order



Topic is order

***A
xx
mth